MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL,
8-BIT IDENTITY COMPARATOR
MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, 8-Bit Identity Comparator, Logic Gate microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	8-Bit Identity Comparator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
R	D-8 (20-lead, 1/4" x 1 1/16") dual-in-line package.
S	F-9 (20-lead, 1/4" x 1/2") flat package.
2	C-2 (20-terminal, .350" x .350"), square chip carrier.

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	176 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}) 2/:	
Cases R, S	(See MIL-M-38510, appendix C)
Case 2	0.08°C/W
Junction temperature (T_J)	+175°C 3/

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (VCC)	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 4/	
Low logic level	33 maximum
High logic level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.2.2 Truth table. The truth table shall be as specified on figure 2.

3.2.3 Timing diagrams. The timing diagrams shall be as specified on figure 3.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

4/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA, V_{IL} = 0.8 V	01	2.5		V
Low-level ouput voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 20 mA, V_{IH} = 2.0 V	01		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, IIN = -18 mA, T_C = +25°C	01		-1.2	V
High-level input current	I_{IH1} I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V V_{CC} = 5.5 V, V_{IN} = 7.0 V	01 01		20 100	µA µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	01	-.03	-.60	mA
Short circuit output current 1/	I_{OS}	V_{CC} = 5.5 V	01	-60	-150	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01		32	mA
Propagation delay time, low to high level (In Phase)	t_{PLH1} t_{PLH2} t_{PLH3}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 500 ±5%	01 01 01	1.0 1.0 1.0	15.0 15.0 8.5	ns ns ns
Propagation delay time, low to high level (Out of Phase)	t_{PLH4} t_{PLH5}		01 01	1.0 1.0	15.0 15.0	ns ns
Propagation delay time, high to low level (In Phase)	t_{PHL1} t_{PHL2} t_{PHL3}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 500 ±5%	01 01 01	1.0 1.0 1.0	15.0 15.0 13.5	ns ns ns
Propagation delay time, high to low level (Out of Phase)	t_{PHL4} t_{PHL5}		01 01	1.0 1.0	15.0 15.0	ns ns

1/ Not more than one output should be shorted at a time

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 8 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, 6, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to group C inspection for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

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c. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

(1) Test condition D or E, using the curcuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3, 9	1,2,3, 7,9
Group A test requirements (method 5005)	1,2,3,7, 9,10,11	1,2,3,7 9
Group B test requirements (method 5005) subgroup 5	1,2,3,7, 9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C inspections	N/A	10, 11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

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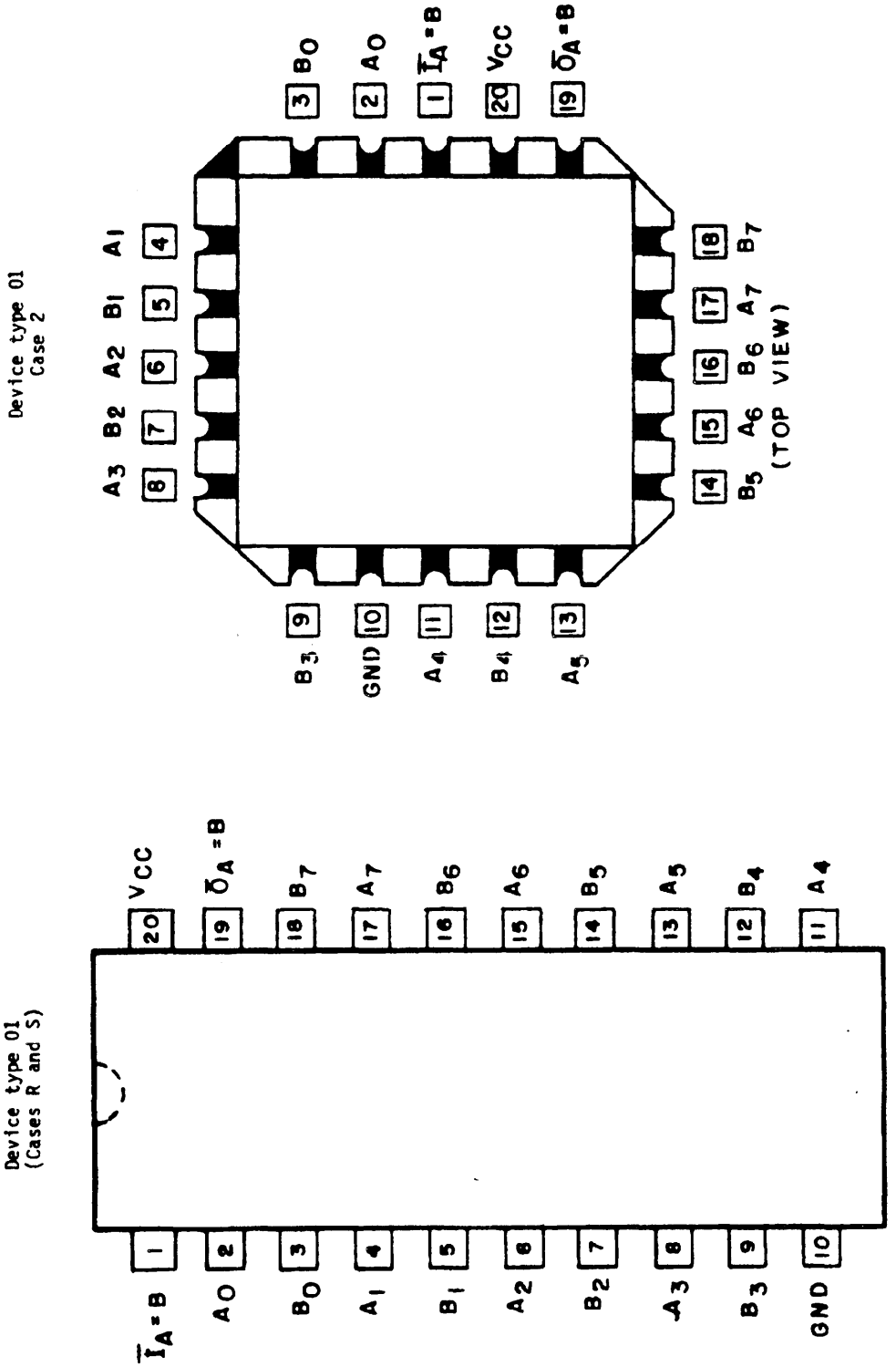


FIGURE 1. Terminal connections and logic diagram (top view).

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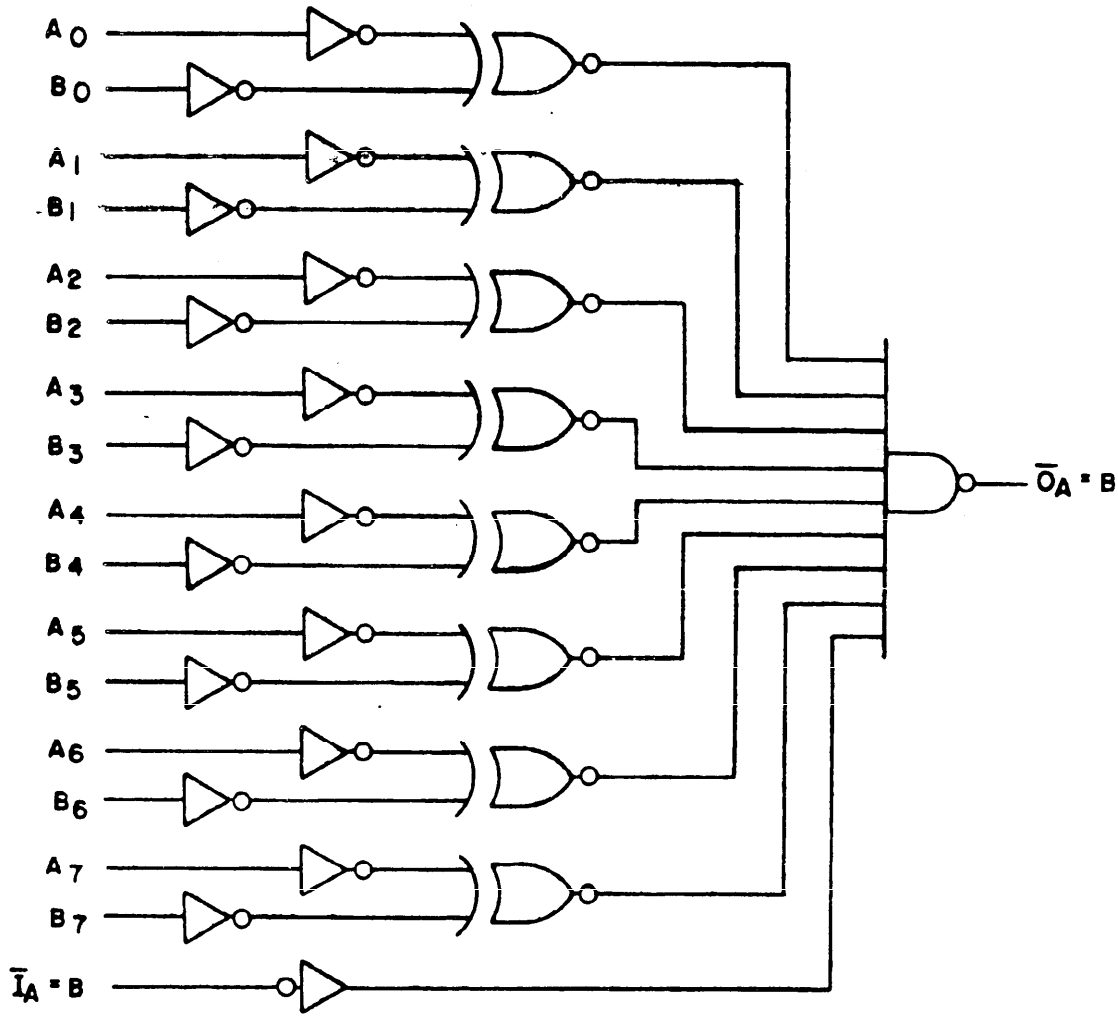


FIGURE 1. Terminal connections and logic diagram (top view) - Continued.

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Truth Table		
Inputs		Output
$\overline{I}_A = B$	A,B	$\overline{O}_A = B$
L	A = B	L
L	A ≠ B	H
H	A = B	H
H	A ≠ B	H

H = HIGH Voltage Level
L = LOW Voltage Level
$A_0 = B_0$, $A_1 = B_1$, $A_2 = B_2$, etc

FIGURE 2. Truth table.

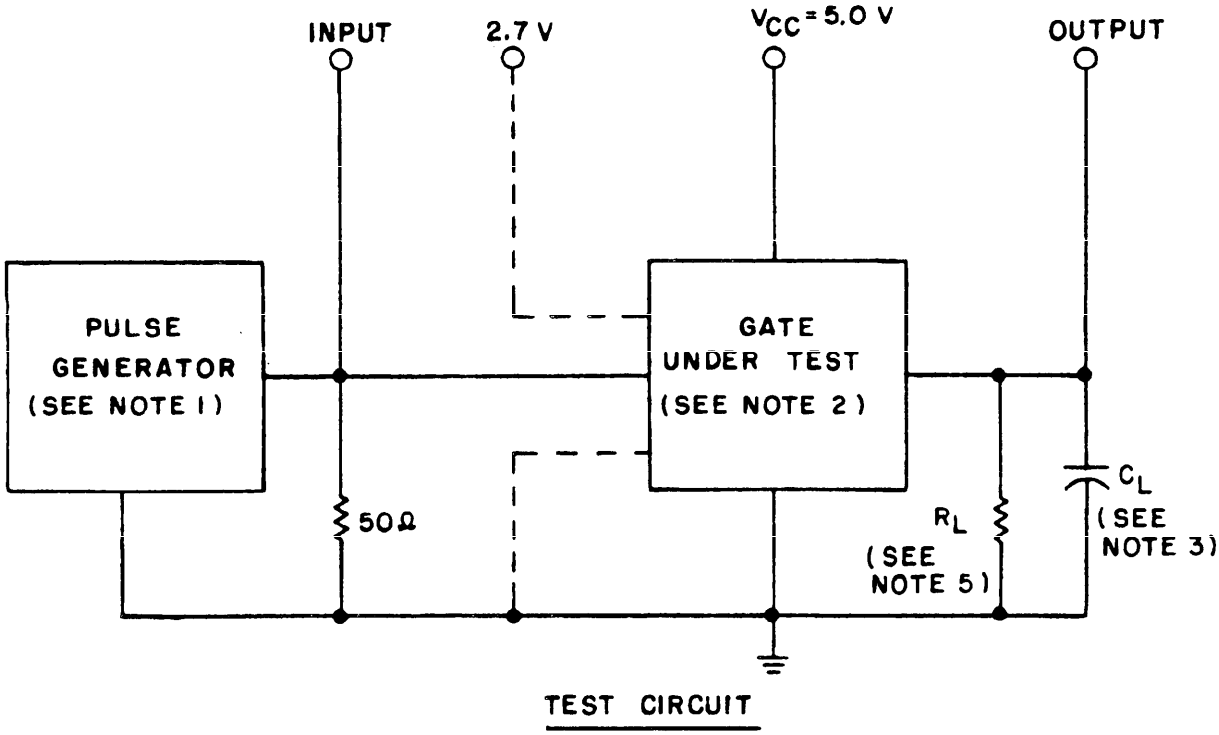
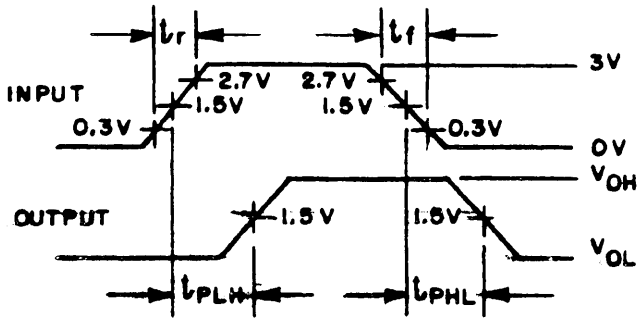


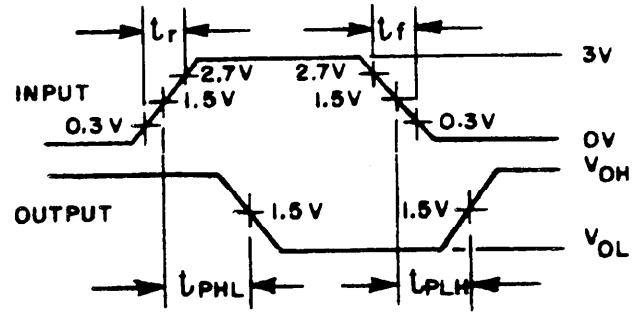
FIGURE 3. Switching time test circuit and waveforms.

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Waveform for in-phase propagation delays



Waveform for out-of-phase propagation delays

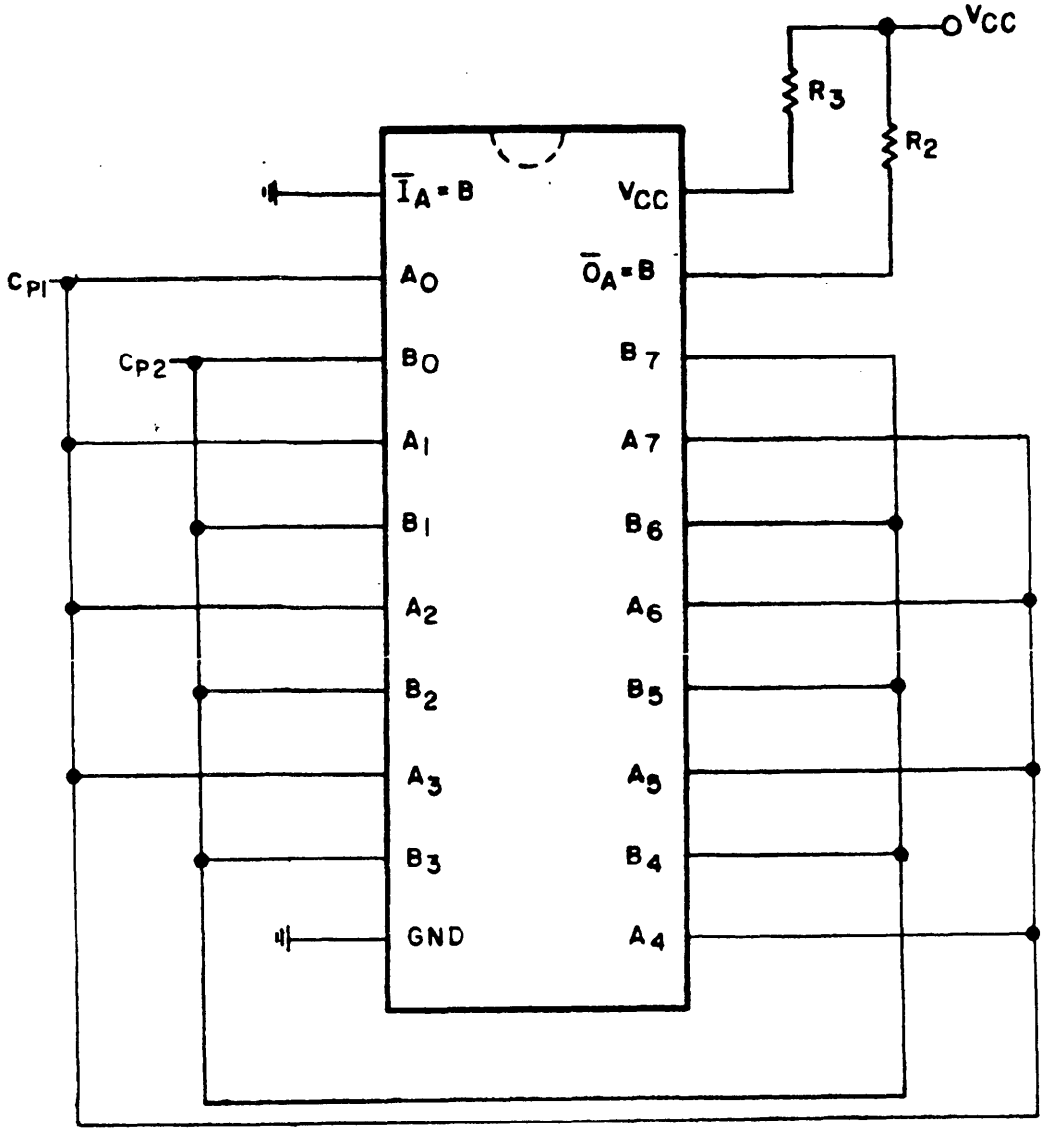


NOTES:
1. Pulse generator has the following characteristics: $t_r = t_f \leq 2.5$ ns, $t_p = 200$ ns. $Z_{out} \approx 50\Omega$ and PRR $\leq$ 1 MHz.
2. Inputs not under test are at 2.7 V or GND as specified in table III.
3. $C_L = 50$ pF ±10% including scope, probe, wiring, and stray capacitance, without package in test fixture.
4. Voltage measurements are to be made with respect to network ground terminal.
5. $R_L = 500\Omega$ ±5%.

FIGURE 3. Switching time test circuit and waveforms - Continued.

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Device type 01
Cases R, S and 2



NOTES:
1. R_1 = 51Ω ±5%.
   R_2 = 270Ω ±5%.
2. V_{CC} and R_3 shall be chosen to insure a 5.5 V minimum is present at device V_{CC} terminal.
3. $C.P._1$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%; C_{P2} = 1/2 C_{P1} V_{IL} = -0.5 V min to 0.8 V max; V_{IH} = 2.0 V min to 5.5 V max.

FIGURE 4. Burn-in and life test circuit.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 TA=8	2 A0	3 B0	4 A1	5 B1	6 A2	7 B2	8 A3	9 B3	10 GND	11 A4	12 B4	13 A5	14 B5	15 A6	16 B6	17 A7	18 B7	19 IOA=B	20 VCC	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0	0	5.5	0	5.5	0	5.5	0	5.5	GND	0	5.5V	0	5.5V	0	5.5V	0	5.5V	-1mA	4.5V	OA=B	2.5V		V
			2	0	0.8	2.0	0	0	0	0	0	0		0	0	0	0	0	0	0	0						
			3	0	2.0	0.8	0	0	0	0	0	0		0	0	0	0	0	0	0	0						
			4	0	0	0	0.8	2.0	0	0	0	0		0	0	0	0	0	0	0	0						
			5	0	0	0	2.0	0.8	0	0	0	0		0	0	0	0	0	0	0	0						
			6	0	0	0	0	0	0.8	2.0	0	0		0	0	0	0	0	0	0	0						
			7	0	0	0	0	0	2.0	0.8	0	0		0	0	0	0	0	0	0	0						
			8	0	0	0	0	0	0	0	0.8	2.0		0	0	0	0	0	0	0	0						
			9	0	0	0	0	0	0	0	2.0	0.8		0	0	0	0	0	0	0	0						
			10	0	0	0	0	0	0	0	0	0		0.8V	0.8V	0	0	0	0	0	0						
			11	0	0	0	0	0	0	0	0	0		2.0V	2.0V	0	0	0	0	0	0						
			12	0	0	0	0	0	0	0	0	0		0	0	0.8V	2.0V	0	0	0	0						
			13	0	0	0	0	0	0	0	0	0		0	0	2.0V	0.8V	0	0	0	0						
			14	0	0	0	0	0	0	0	0	0		0	0	0	0	0.8V	0	0.8V	2.0V						
			15	0	0	0	0	0	0	0	0	0		0	0	0	0	2.0V	0	2.0V	0.8V						
			16	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0.8V	0	0						
			17	0	0	0	0	0	0	0	0	0		0	0	0	0	0	2.0V	0	0						
	V_{OL}	3007	18	0.8	0	0	0	0	0	0	0	0	GND	0	0	0	0	0	0	0	20mA	4.5V	OA=B		0.5V	V
	V_{IC}		19	-18mA																		4.5V	$I_{A=B}$	-1.2		V
			20		-18mA																			A0			
			21			-18mA																		B0			
			22				-18mA																	A1			
			23					-18mA																B1			
			24						-18mA															A2			
			25							-18mA														B2			
			26								-18mA													A3			
			27									-18mA												B3			
			28											-18mA										A4			
			29												-18mA									B4			
			30													-18mA								A5			
			31														-18mA							B5			
			32															-18mA						A6			
			33																-18mA					B6			
			34																	-18mA				A7			
			35																		-18mA			B7			
	I_{IH1}	3010	36	2.7V	2.7V								GND									5.5V	$I_{A=B}$	20		uA
			37			2.7V																		A0			
			38				2.7V																	B0			
			39					2.7V																A1			
			40						2.7V															B1			
			41							2.7V														A2			
			42								2.7V													B2			
			43									2.7V												A3			
			44											2.7V										B3			
			45												2.7V									A4			
			46													2.7V								B4			
			47														2.7V							A5			
			48															2.7V						B5			
			49																2.7V					A6			
			50																	2.7V				B6			
			51																		2.7V			A7			
			52																					B7			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{TA=B}$	A_0	B_0	A_1	B_1	A_2	B_2	A_3	B_3	GND	A_4	B_4	A_5	B_5	A_6	B_6	A_7	B_7	$\overline{OA=B}$	VCC				
1 T_C = +25°C	I_{IH2}	3010	53	7.0V									GND									5.5V	$\overline{T_{A=B}}$		100	uA
	"	"	54		7.0V								"									"	A_0		"	"
	"	"	55			7.0V							"									"	B_0		"	"
	"	"	56				7.0V						"									"	A_1		"	"
	"	"	57					7.0V					"									"	B_1		"	"
	"	"	58						7.0V				"									"	A_2		"	"
	"	"	59							7.0V			"									"	B_2		"	"
	"	"	60								7.0V		"									"	A_3		"	"
	"	"	61									7.0V	"									"	B_3		"	"
	"	"	62										"	7.0V								"	A_4		"	"
	"	"	63										"		7.0V							"	B_4		"	"
	"	"	64										"			7.0V						"	A_5		"	"
	"	"	65										"				7.0V					"	B_5		"	"
	"	"	66										"					7.0V				"	A_6		"	"
	"	"	67										"						7.0V			"	B_6		"	"
	"	"	68										"							7.0V		"	A_7		"	"
	"	"	69										"								7.0V	"	"	B_7		"	"
	I_{IL}	3009	70	0.5V									GND									5.5V	$\overline{T_{A=B}}$	1/	1/	mA
	"	"	71		0.5V								"									"	A_0	"	"	"
	"	"	72			0.5V							"									"	B_0	"	"	"
	"	"	73				0.5V						"									"	A_1	"	"	"
	"	"	74					0.5V					"									"	B_1	"	"	"
	"	"	75						0.5V				"									"	A_2	"	"	"
	"	"	76							0.5V			"									"	B_2	"	"	"
	"	"	77								0.5V		"									"	A_3	"	"	"
	"	"	78									0.5V	"									"	B_3	"	"	"
	"	"	79										"	0.5V								"	A_4	"	"	"
	"	"	80										"		0.5V							"	B_4	"	"	"
	"	"	81										"			0.5V						"	A_5	"	"	"
	"	"	82										"				0.5V					"	B_5	"	"	"
	"	"	83										"					0.5V				"	A_6	"	"	"
	"	"	84										"						0.5V			"	B_6	"	"	"
	"	"	85										"							0.5V		"	A_7	"	"	"
	"	"	86										"								0.5V	"	"	B_7	"	"	"
	I_{OS}	3011	87	5.5V									GND									0	5.5V	$\overline{OA=B}$	-60	-150	mA
	I_{CC}	3005	88	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V	GND	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V		5.5V	V_{CC}		32	mA
2	Same tests, terminal condition, and limits as for subgroup 1, except TC = +125° and V_{IC} tests are omitted.																										
3	Same tests, terminal condition, and limits as for subgroup 1, except TC = -55° and V_{IC} tests are omitted.																										
7 T_C = +25°C	Truth Table Test	3014	89	B	B	B	B	B	B	B	B	B	GND	B	B	B	B	B	B	B	B	L	5.0V	All Outputs			
	"	"	90	B	A	A	A	A	A	A	A	A	"	A	A	A	A	A	A	A	A	L	"	"			
	"	"	91	A	A	A	A	A	A	A	A	A	"	A	A	A	A	A	A	A	A	H	"	"			
	"	"	92	A	B	B	B	B	B	B	B	B	"	B	B	B	B	B	B	B	B	H	"	"			
	"	"	93	B	A	B	B	B	B	B	B	B	"	B	B	B	B	B	B	B	B	H	"	"			
	"	"	94	B	B	A	B	B	B	B	B	B	"	B	B	B	B	B	B	B	B	H	"	"			
	"	"	95	B	B	B	A	B	B	B	B	B	"	B	B	B	B	B	B	B	B	H	"	"			
	"	"	96	B	B	B	B	A	B	B	B	B	"	B	B	B	B	B	B	B	B	H	"	"			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 TA=B	2 A_0	3 B_0	4 A_1	5 B_1	6 A_2	7 B_2	8 A_3	9 B_3	10 GND	11 A_4	12 B_4	13 A_5	14 B_5	15 A_6	17 A_7	18 B_7	19 $\overline{OA}$=B	20 VCC	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = +25°C	Truth Table Test	3014	97	B	B	B	B	B	A	B	B	B	"	B	B	B	B	B	B	B	H	5.0 V	All outputs		
	"	"	98	B	B	B	B	B	B	A	B	B	"	B	B	B	B	B	B	B	H	"	"		
	"	"	99	B	B	B	B	B	B	B	A	B	"	B	B	B	B	B	B	B	H	"	"		
	"	"	100	B	B	B	B	B	B	B	B	A	"	B	B	B	B	B	B	B	H	"	"		
	"	"	101	B	B	B	B	B	B	B	B	B	"	A	B	B	B	B	B	B	H	"	"		
	"	"	102	B	B	B	B	B	B	B	B	B	"	B	A	B	B	B	B	B	H	"	"		
	"	"	103	B	B	B	B	B	B	B	B	B	"	B	B	A	B	B	B	B	H	"	"		
	"	"	104	B	B	B	B	B	B	B	B	B	"	B	B	B	A	B	B	B	H	"	"		
	"	"	105	B	B	B	B	B	B	B	B	B	"	B	B	B	B	A	B	B	H	"	"		
	"	"	106	B	B	B	B	B	B	B	B	B	"	B	B	B	B	B	B	B	H	"	"		
	"	"	107	B	B	B	B	B	B	B	B	B	"	B	B	B	B	B	A	B	H	"	"		
	"	"	108	B	B	B	B	B	B	B	B	B	"	B	B	B	B	B	B	A	H	"	"		
9 T_C = +25°C	t_{PLH1}	3003	109	GND	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	A0to$\overline{OA}$-B	1.5	10.0	ns
	"	"	110	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A1to$\overline{OA}$-B	"	"	"
	"	"	111	"	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2to$\overline{OA}$-B	"	"	"
	"	"	112	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	A3to$\overline{OA}$-B	"	"	"
	"	"	113	"	"	"	"	"	"	"	GND	"	"	IN	"	"	"	"	"	"	"	"	"	A4to$\overline{OA}$-B	"	"	"
	"	"	114	"	"	"	"	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	A5to$\overline{OA}$-B	"	"	"
	"	"	115	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	A6to$\overline{OA}$-B	"	"	"
	"	"	116	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	IN	"	"	"	"	A7to$\overline{OA}$-B	"	"	"
	t_{PLH2}	"	117	GND	GND	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	B0to$\overline{OA}$-B	1.5	10.0	ns
	"	"	118	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1to$\overline{OA}$-B	"	"	"
	"	"	119	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	B2to$\overline{OA}$-B	"	"	"
	"	"	120	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	B3to$\overline{OA}$-B	"	"	"
	"	"	121	"	"	"	"	"	"	"	"	GND	"	"	IN	"	"	"	"	"	"	"	"	B4to$\overline{OA}$-B	"	"	"
	"	"	122	"	"	"	"	"	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	B5to$\overline{OA}$-B	"	"	"
	"	"	123	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	"	"	"	B6to$\overline{OA}$-B	"	"	"
	"	"	124	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	IN	"	"	"	B7to$\overline{OA}$-B	"	"	"
	t_{PLH3}	"	125	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	$\overline{TA}$=BtoA-B	1.5	6.5	
	t_{PLH4}	"	126	GND	IN	2.7V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	A0to$\overline{OA}$-B	1.5	10.0	ns
	"	"	127	"	GND	GND	IN	2.7V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A1to$\overline{OA}$-B	"	"	"
	"	"	128	"	"	"	GND	GND	IN	2.7V	"	"	"	"	"	"	"	"	"	"	"	"	"	A2to$\overline{OA}$-B	"	"	"
	"	"	129	"	"	"	"	"	GND	GND	IN	2.7V	"	"	"	"	"	"	"	"	"	"	"	A3to$\overline{OA}$-B	"	"	"
	"	"	130	"	"	"	"	"	"	"	GND	GND	"	IN	2.7V	"	"	"	"	"	"	"	"	A4to$\overline{OA}$-B	"	"	"
	"	"	131	"	"	"	"	"	"	"	"	"	"	GND	GND	IN	2.7V	"	"	"	"	"	"	A5to$\overline{OA}$-B	"	"	"
	"	"	132	"	"	"	"	"	"	"	"	"	"	"	"	GND	GND	IN	"	"	"	"	"	A6to$\overline{OA}$-B	"	"	"
	"	"	133	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	IN	2.7V	"	"	"	A7to$\overline{OA}$-B	"	"	"
	t_{PLH5}	"	134	GND	2.7V	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	B0to$\overline{OA}$-B	1.5	10.0	ns
	"	"	135	"	GND	GND	2.7V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1to$\overline{OA}$-B	"	"	"
	"	"	136	"	"	"	GND	GND	2.7V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	B2to$\overline{OA}$-B	"	"	"
	"	"	137	"	"	"	"	"	GND	GND	2.7V	IN	"	"	"	"	"	"	"	"	"	"	"	B3to$\overline{OA}$-B	"	"	"
	"	"	138	"	"	"	"	"	"	"	GND	GND	"	2.7V	IN	"	"	"	"	"	"	"	"	B4to$\overline{OA}$-B	"	"	"
	"	"	139	"	"	"	"	"	"	"	"	"	"	GND	GND	2.7V	IN	"	"	"	"	"	"	B5to$\overline{OA}$-B	"	"	"
	"	"	140	"	"	"	"	"	"	"	"	"	"	"	"	GND	GND	2.7V	"	"	"	"	"	B6to$\overline{OA}$-B	"	"	"
	"	"	141	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	2.7V	IN	"	"	"	B7to$\overline{OA}$-B	"	"	"
	t_{PHL1}	3003	142	GND	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	A0to$\overline{OA}$-B	1.5	10.0	ns
	"	"	143	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A1to$\overline{OA}$-B	"	"	"
	"	"	144	"	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2to$\overline{OA}$-B	"	"	"
	"	"	145	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	A3to$\overline{OA}$-B	"	"	"
	"	"	146	"	"	"	"	"	"	"	GND	"	"	IN	"	"	"	"	"	"	"	"	"	A4to$\overline{OA}$-B	"	"	"
	"	"	147	"	"	"	"	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	A5to$\overline{OA}$-B	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 TA=B	2 A_0	3 B_0	4 A_1	5 B_1	6 A_2	7 B_2	8 A_3	9 B_3	10 GND	11 A_4	12 B_4	13 A_5	14 B_5	15 A_6	16 B_6	17 A_7	18 B_7	19 $\overline{OA=B}$	20 VCC	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL1}	3003	148	"	"	"	"	"	"	"	"	"	"	GND	GND	GND	GND	IN	GND	GND	GND	OUT	5.0 V	A6to$\overline{OA=B}$	1.5	10.0	ns
	"	"	149	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	IN	"	"	"	A7to$\overline{OA=B}$	"	"	"
	t_{PHL2}	"	150	GND	GND	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	B0to$\overline{OA=B}$	1.5	10.0	ns
	"	"	151	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1to$\overline{OA=B}$	"	"	"
	"	"	152	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	B2to$\overline{OA=B}$	"	"	"
	"	"	153	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"	B3to$\overline{OA=B}$	"	"	"
	"	"	154	"	"	"	"	"	"	"	"	GND	"	"	IN	"	"	"	"	"	"	"	"	B4to$\overline{OA=B}$	"	"	"
	"	"	155	"	"	"	"	"	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	B5to$\overline{OA=B}$	"	"	"
	"	"	156	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	IN	"	"	"	"	B6to$\overline{OA=B}$	"	"	"
	"	"	157	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	IN	"	"	B7to$\overline{OA=B}$	"	"	"
	t_{PHL3}	"	158	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0 V	$\overline{Ta=B}$to$\overline{OA=B}$	1.5	9.0	ns
	t_{PHL4}	"	159	GND	IN	2.7V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	A0to$\overline{OA=B}$	1.5	10.0	ns
	"	"	160	"	GND	GND	IN	2.7V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A1to$\overline{OA=B}$	"	"	"
	"	"	161	"	"	"	GND	GND	IN	2.7V	"	"	"	"	"	"	"	"	"	"	"	"	"	A2to$\overline{OA=B}$	"	"	"
	"	"	162	"	"	"	"	"	GND	GND	IN	2.7V	"	"	"	"	"	"	"	"	"	"	"	A3to$\overline{OA=B}$	"	"	"
	"	"	163	"	"	"	"	"	"	"	GND	GND	"	IN	2.7V	"	"	"	"	"	"	"	"	A4to$\overline{OA=B}$	"	"	"
	"	"	164	"	"	"	"	"	"	"	"	"	"	GND	GND	IN	2.7V	"	"	"	"	"	"	A5to$\overline{OA=B}$	"	"	"
	"	"	165	"	"	"	"	"	"	"	"	"	"	"	"	GND	GND	IN	2.7V	"	"	"	"	A6to$\overline{OA=B}$	"	"	"
	"	"	166	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	GND	IN	2.7V	"	"	A7to$\overline{OA=B}$	"	"	"
	t_{PHL5}	"	167	GND	2.7V	IN	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	5.0V	B0to$\overline{OA=B}$	1.5	10.0	ns
	"	"	168	"	GND	GND	2.7V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1to$\overline{OA=B}$	"	"	"
	"	"	169	"	"	"	GND	GND	2.7V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	B2to$\overline{OA=B}$	"	"	"
	"	"	170	"	"	"	"	"	GND	GND	2.7V	IN	"	"	"	"	"	"	"	"	"	"	"	B3to$\overline{OA=B}$	"	"	"
	"	"	171	"	"	"	"	"	"	"	GND	GND	"	2.7V	IN	"	"	"	"	"	"	"	"	B4to$\overline{OA=B}$	"	"	"
	"	"	172	"	"	"	"	"	"	"	"	"	"	GND	GND	2.7V	IN	"	"	"	"	"	"	B5to$\overline{OA=B}$	"	"	"
	"	"	173	"	"	"	"	"	"	"	"	"	"	"	"	GND	GND	2.7V	IN	"	"	"	"	B6to$\overline{OA=B}$	"	"	"
	"	"	174	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	GND	2.7V	IN	"	"	B7to$\overline{OA=B}$	"	"	"
10	Same test and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																									
11	Same test and terminal conditions as subgroup 9, except T_C = -55°C and use limits from table I.																									

1/

Test	Min/Max limits in mA for circuits: A	B	C
I_{IL}	-.25/-.60	-.03/-.60	

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4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - Ground zero voltage potential.
I_{IN} - - - - - - - - - - - - - - - Current flowing into an input terminal.
V_{IN} - - - - - - - - - - - - - - - Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F521

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6.6 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit				
	A	B	C		
	Fairchild Semiconductor	Motorola	Signetics		
01	X	X			

Custodians:
Air Force - 17
Navy - EC
Army - ER

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0750)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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